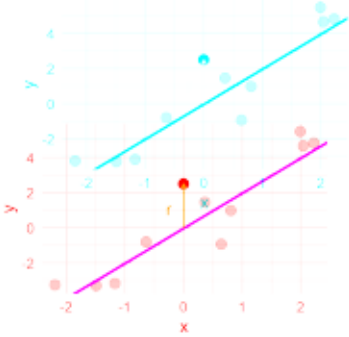
# Advanced Risk Minimization

## Pseudo-Residuals and Gradient Descent




**Learning goals**

- Know the concept of pseudo-residuals
- Understand the relationship between pseudo-residuals and gradient descent

**Learning goals**

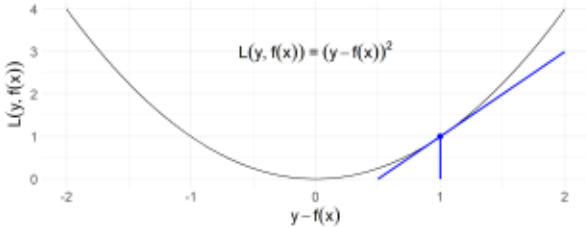
- Know the concept of pseudo-residuals
- Understand the relationship between pseudo-residuals and gradient descent

# PSEUDO-RESIDUALS

- In regression, residuals are defined as $r := y - f(\mathbf{x})$.
- We further define **pseudo-residuals** as the negative first derivatives of loss functions w.r.t. $f(\mathbf{x})$

$$\tilde{r} \quad := \quad -\frac{\partial L\left(y, f(\mathbf{x})\right)}{\partial f(\mathbf{x})}.$$

- This definition also holds for score / probability based classifiers.
- Note that $\tilde{r}$ depends on $y$ and $f(\mathbf{x})$ and $L$.



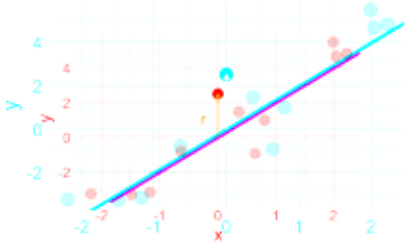
# BEST POINT-WISE UPDATE

Assume we have (partially) fitted a model $f(\mathbf{x})$ to data $\mathcal{D}$.

Assume we could update $f(\mathbf{x})$ point-wise as we like. Under squared loss, for a fixed $\mathbf{x} \in \mathcal{X}$, the best point-wise update is the direction of the residual $r = y - f(\mathbf{x})$

$$f(\mathbf{x}) \leftarrow f(\mathbf{x}) + r$$

$$f(\mathbf{x}) \leftarrow f(\mathbf{x}) + r$$




**Caveat**: In GB, we don't do point-wise updates but "smoothly distort" $f$ so we generalize

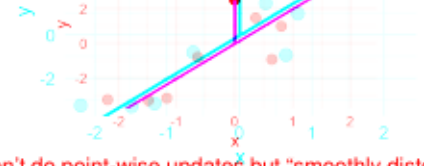
# BEST POINT-WISE UPDATE

Assume we have (partially) fitted a model $f(\mathbf{x})$ to data $\mathcal{D}$.

Assume we could update $f(\mathbf{x})$ point-wise as we like. Under squared loss, for a fixed $\mathbf{x} \in \mathcal{X}$, the best point-wise update is the direction of the residual $r = y - f(\mathbf{x})$

$$f(\mathbf{x}) \leftarrow f(\mathbf{x}) + r$$

The point-wise error at this specific $\mathbf{x}$ becomes 0.




**Caveat**: In GB, we don't do point-wise updates but "smoothly distort" $f$ so we generalize

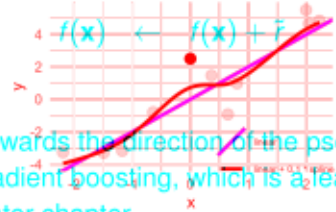
# BEST POINT-WISE UPDATE

Assume we have (partially) fitted a model $f(\mathbf{x})$ to data $\mathcal{D}$.

Assume we could update $f(\mathbf{x})$ point-wise as we like. Under squared loss, for a fixed $\mathbf{x} \in \mathcal{X}$, the best point-wise update is the direction of the residual $r = y - f(\mathbf{x})$

which is the direction of the pseudo-residual

$$f(\mathbf{x}) \leftarrow f(\mathbf{x}) + r$$




**Caveat**: In GB, we don't do point-wise updates but "smoothly distort" $f$ so we generalize

# APPROXIMATE BEST POINT-WISE UPDATE

When applying gradient descent (GD) to compute a point-wise update of $f(\mathbf{x})$, we would go a step into the direction of the negative gradient

$$f(\mathbf{x}) \leftarrow f(\mathbf{x}) - \frac{\partial L(y, f(\mathbf{x}))}{\partial f(\mathbf{x})}.$$

- This can be seen as approximating the unexplained information (measured by the loss) through a model update.
- Using the chain rule:

$$\nabla_\theta \mathcal{R}_{\text{emp}}(\theta) = \sum_{i=1}^{n} \frac{\partial L(y^{(i)}, f)}{\partial f}\bigg|_{f=f\left(\mathbf{x}^{(i)} \mid \theta\right)} \cdot \nabla_\theta f\left(\mathbf{x}^{(i)} \mid \theta\right)$$

Iteratively stepping towards the direction of the pseudo-residuals is the underlying idea of gradient boosting, which is a learning algorithm that will be covered in a later chapter.

- Hence the update is determined by a loss-optimal directional change of the model output and a loss-independent derivate of $f$. This is a very flexible, nearly loss-independent formulation of GD.